

1 The students table

1 List all distinct departments

SQL :

SELECT DISTINCT department
FROM students;

OUTPUT :

Results

department
IT
HR
Finance

2 Average age # of students per department

SQL :

SELECT department, (AVG)age AS avg_age
FROM students
GROUP BY department;

Results

department	avg_age
IT	20.5
HR	22
Finance	23

3 Show departments with more than one student

SELECT department, COUNT AS student_count
FROM students
GROUP BY department
HAVING COUNT > 1;

Results

department	student_count
IT	2
HR	2

④ Get all students whose age is between 21 and 23

SELECT *

FROM students

WHERE age BETWEEN 21 AND 23;

Results

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ List all the students in the IT or HR department who are older than 21

SELECT student_id,
name,
age,
department

FROM students

WHERE department IN ('IT', 'HR')

AND age > 21;

Results

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

(8) Show the top 3 courses by credits in descending order

```
SELECT course_id,  
       course_name,  
       credits  
FROM   courses  
ORDER BY credits DESC  
LIMIT 3;
```

Results

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

(9) Maximum, minimum and average grade

```
SELECT MAX(grade) AS max_grade,  
       MIN(grade) AS min_grade,  
       AVG(grade) AS avg_grade  
FROM   enrollments;
```

Result

max_grade	min_grade	avg_grade
90	78	84.6

(10) Count how many enrollments exist per course (5)

```
SELECT course_id,  
       COUNT(*) AS enrollment_count  
FROM enrollments  
GROUP BY course_id;
```

Results

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(11) Find total salary and total bonus per department

```
SELECT department,  
       SUM(salary) AS total_salary,  
       SUM(bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```

Results

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

(6)

(12) Departments with average salary above 55000

```

SELECT department,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;

```

Results

department	avg_salary
IT	61000
Finance	70000

(13) Employees whose salary plus bonus is greater than 60000

```

SELECT employee_id,
       name,
       salary,
       bonus
       (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;

```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

(14) Total and average budget per department
(average above 70 000)

(7)

```
SELECT department,  
       sum(budget) AS total-budget,  
       Avg (budget) AS avg-budget  
FROM projects  
GROUP BY department  
HAVING AVG (budget) > 70000;
```

Results

department	total - budget	avg - budget
IT	270000	135000
Finance	80000	80000

(15) Projects with budgets between 50000 and 120 000
excluding Marketing.

```
SELECT project-id,  
       project-name,  
       department,  
       budget  
FROM projects  
WHERE budget BETWEEN 50000 and 120 000  
AND department <> 'Marketing';
```

Results

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000